

SABA KHARABADZE, PHD

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Pisa, Tuscany, Italy

SUMMARY

ML Research Engineer / Research Scientist with 7+ years building ML-enabled scientific and production systems. Experienced in GPU/distributed workflows, large-scale data generation and processing, HPC environments (GPU nodes, job scheduling), and C/C++ + Python ML pipelines (PyTorch/JAX).

SKILLS

Programming	Python (PyTorch, JAX, NumPy, SciPy, pandas), C/C++, SQL, Bash, Git
Systems/Infra	Docker, Linux, HPC (GPU nodes, schedulers), CUDA, multi-GPU training, distributed training
Scientific/Domain	DFT (VASP, Gaussian), materials modeling, numerical simulation
Languages	English (native/bilingual), Georgian (native/bilingual), Russian (proficient)

RESEARCH ENGINEERING EXPERIENCE

ML Research Engineer (Research Scientist) - *CNIT RaSS Lab* - *Pisa, Italy* April 2025 - Present

- Build ML-based radar signal processing workflows for target detection.
- Develop simulation frameworks for coverage analysis of mono- and multistatic radar systems.
- Implement reproducible experimentation code and analysis pipelines for model evaluation and system-level studies.

Physics AI Model Validation Expert - *Handshake MOVE Program* - *Remote* August 2025 - Present

- Design physics reasoning problems and reference solutions to stress-test AI model capabilities.
- Evaluate model responses and write targeted feedback to improve reasoning and physics correctness.

PhD Candidate - *Binghamton University (SUNY)* - *Binghamton, NY* August 2018 - August 2024

- Built end-to-end pipelines for ML-accelerated atomistic simulation: data generation, model training, and high-throughput search.
- Scaled evolutionary search + ML potential workflows to 1M+ structures and 1TB energy/force data using HPC clusters.
- Developed Python tooling to transform raw simulation outputs into physical/electrochemical/structural properties used for discovery decisions.
- Implemented and validated an NPT barostat module in C for in-house molecular dynamics software (MAISE); achieved thermal expansion coefficients within 10% of experiment.
- Administered two university HPC clusters (1000+ CPU cores plus GPU nodes): installed/maintained compilers, CUDA, MPI, PyTorch stack, and scientific codes; enabled multi-node and multi-GPU use.

SELECTED PROJECT HIGHLIGHTS

- **ML-accelerated discovery of Li-Sn phases:** created DFT reference data (VASP), trained/used a neural network potential (C), and executed high-throughput evolutionary searches; identified **8 new stable structures** and validated stability with phonons.
- **Thermodynamic stability of Li-B-C compounds:** computed chemical potentials and performed quasi-harmonic phonon analysis to study stability across volumes and temperatures; published results in PCCP.
- **Engineering for reproducibility:** built scripts, Python tooling and deployed documentation website (Sphinx) for simulation workflows; maintained a public documentation site for MAISE.

EDUCATION

PhD in Physics , Binghamton University, State University of New York Dissertation: Machine Learning and <i>ab initio</i> insights into the design of lithium-based materials.	May 2024
B.S. in Physics , Free University of Tbilisi Minor: Computer Science and Mathematics	May 2017

SELECTED PUBLICATIONS

- Kharabadze, S. *et al.* (2022) Prediction of stable Li-Sn compounds: boosting ab initio searches with neural network potentials. *npj Comput Mater.*
- Kharabadze, S. *et al.* (2023) Thermodynamic stability of Li-B-C compounds from first principles. *Phys. Chem. Chem. Phys.*
- Hajinazar, S. *et al.* (2021) MAISE: Construction of neural network interatomic models and evolutionary structure optimization. *Comput. Phys. Commun.*

ADDITIONAL INDUSTRY EXPERIENCE

Founding Member, Data Science / ML - <i>Keepers</i> - https://kprs.pro - <i>Remote</i>	Dec 2024 – Present
- Build ETL pipelines to ingest exchange data into internal databases using Python, SQLAlchemy, and SQL; implement data validation and cleaning in pandas. - Develop anomaly detection and trend prediction on financial time-series using XGBoost and feature engineering pipelines. - Produce research analyses and monitoring reports to support strategy development and ongoing performance review.	
Back-end Developer - <i>Eleven Wireless</i> - <i>Remote (Portland, OR)</i>	Oct 2017 – Apr 2018
- Built a Python middle layer interfacing SQL databases with ElasticSearch; deployed and maintained reporting service on AWS Linux.	
Junior Business Analyst - <i>TBC Bank</i> - <i>Tbilisi, Georgia</i>	May 2017 – Feb 2018
- Produced system documentation and process diagrams within a Scrum-based innovations team.	

HONORS

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| - Handshake AI Fellow, <i>Handshake MOVE Program</i> | Aug 2025 – Present |
| - Travel grant to present at APS March Meeting, <i>Binghamton University</i> | 2022, 2023 |
| - Bronze Medal, International Physics Olympiad (Tallinn, Estonia) | 2012 |